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Palaeomagnetism of the Upper Carboniferous Strzegom and Karkonosze Granites and the Kudowa Granitoid from the Sudet Mountains, Poland

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Carboniferous granites and granitoids, varying in age from 305 to 281 Ma, from the Sudet mountains (latitude 51N, longitude 16E), Poland, carry dual-polarity magnetic components consisting of an inferred primary high-temperature A-component found in the Strzegom Granite, the Karkonosze Granite with the bordering Karpacz Hornfels and the Kudowa Granitoid and scattered low-stability components of Mesozoic–Tertiary age present in all rocks except the Strzegom Granite. The group-A component defines a palaeomagnetic pole position ($D = 202$, $I = -12$, $\alpha_{95} = 13.0$, $N = 4$, calculated palaeomagnetic pole position 42° S, 346° E) which is in good agreement with palaeomagnetic results from rocks of similar age in western Europe, thus implying that no major tectonic movement has taken place in the Sudets relative to stable Europe since Upper Carboniferous times.

1. Introduction

The Strzegom Granite, the Karkonosze Granite with the bordering Karpacz Hornfels and the Kudowa Granitoid from southern Poland (Fig. 1) form part of the Sudet mountains, situated in the northeastern part of the Variscan belt (Fig. 2) which extends from southern Iberia to northeastern Bohemia (Matte, 1986). To the northeast the Variscan belt is obliquely cut off and offset by transcurrent dextral faults, the Elbe and Tornquist–Teisseyre lines (Arthaud and Matte, 1977), which form the southwestern boundary of the Russian platform.

Tectono-metamorphic and magmatic events within the Variscan belt are considered to have occurred between Upper Devonian (360 Ma) and Lower Permian (280 Ma) times (Matte, 1986). Granite/granitoid emplacement is clearly related in space and time to the Variscan orogeny, as the surrounding Carboniferous basins are devoid of such rocks.

Recent geological and geophysical studies have emphasized the geodynamic history of the Variscan belt, which is characterized by tectonic features, i.e., thrust and nappe structures and wrench faults. These structures were created between 380 and 280 Ma and correspond to a large amount of crustal shortening (> 600 km) (Matte, 1986). Various plate-tectonic models have been proposed to explain the composite tectonic nature of the Variscan belt (Behr et al., 1984; Matte, 1986;

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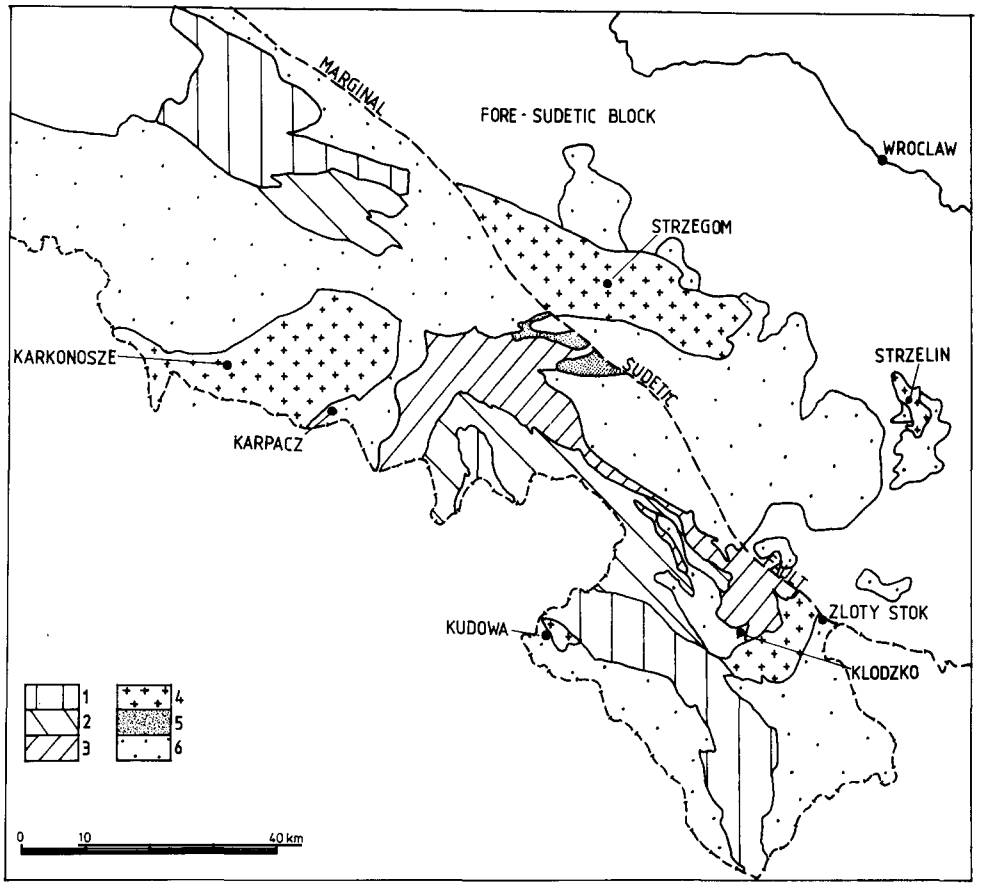


Fig. 1. Geological sketch map of the Sudets simplified from Teisseyre (1980). The rock formations are numbered (1) Upper Cretaceous, (2) Permian, (3) Carboniferous, (4) Variscan granites and granitoids, (5) Devonian and (6) Cambro-Silurian and older.

Perroud, 1986; Franke, 1987), but so far no unique model has emerged. The large width of the belt (1000 km) probably reflects accretion of several microplates situated between the colliding blocks of North Europe and Africa.

Currently, palaeomagnetic investigations have mainly been performed in the western and central parts of the Variscan belt (Bachtadse et al., 1983; Perroud, 1986; Edel, 1987). Bachtadse et al. (1983) showed that the Palaeozoic palaeomagnetic date from the Variscan belt (western and central parts) could be used to interpret the belt in terms of a large-scale shear. They suggested that large megashears with general east–west orientation caused relative anticlockwise rotation of a central European block by $\sim 30°$ of arc with respect to the northern foreland. From a critical review of Devonian and Carboniferous palaeomagnetic results within the Variscan belt, Perroud (1986) found evidence for large rotations. In a study of rocks from the western and central parts of the Variscan belt, Edel (1987) proposed that during the latest Westphalian–Stephanian period, (310–290 Ma), the western European Variscan belt rotated clockwise by 40–50°. Palaeomagnetism is a suitable tool for estimating latitudinal motion and rotations from measurements of inclination and declinations, respectively, and by comparison with adjoining 'stable' continental reference frames. This study reports on palaeomagnetic studies from

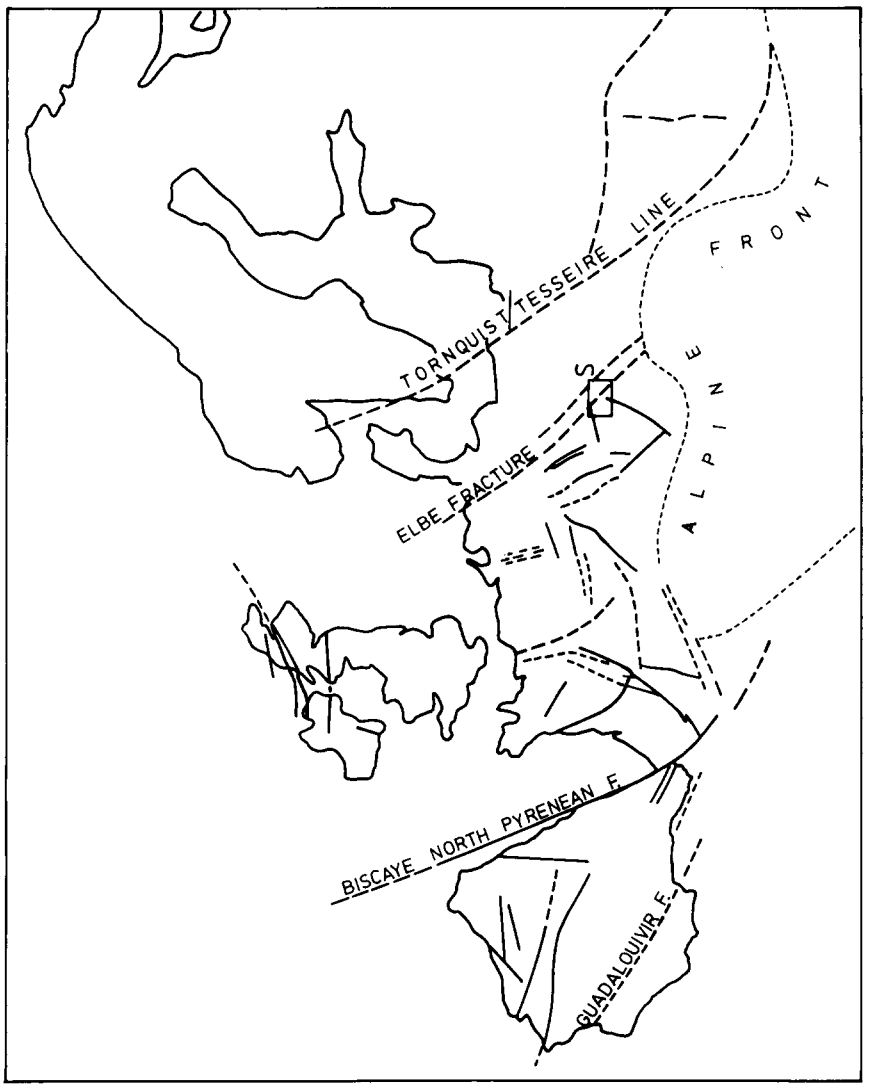


Fig. 2. Map of late Variscan tectonic structures for northwest Europe, simplified from Arthaud and Matte (1977). Heavy continuous and dashed lines represent observed and inferred major fractures, respectively. The location of Fig. 1 is shown by (S).

the Sudet Mountains located in the northeastern part of the Variscan belt.

2. The Sudet granites and granitoids

Granite/granitoid bodies are relatively uniformly distributed within the Sudet area, and are genetically related to the final stage of the Asturian orogenic phase (Oberc, 1977). Palaeomagnetic sampling localities and age determinations are summarized in Table I. According to Ksiazkiewicz et al. (1965), the granite/granitoid intrusions are not tectonically disturbed, thus they are believed to be of post-tectonic origin. All the sampled granites/granitoids are of Upper Carboniferous age. Apart from the igneous bodies specified in Table I, the Strzelin Granite and the Klodzko–Zloty Stok Intrusion (Fig. 1) were also sampled, but these rock formations revealed intensities of magnetization below the noise levels of the magnetometers.

TABLE I

Rock formation sampling details and isotopic ages

Rock formation (latitude, longitude)	Number of:		Age [a] (Ma)	Method
	Sites	Samples		
Strzegom Granite (51° N, 16° E)	9	30	287.7 ± 1.4	Rb–Sr
			300 ± 4	Fission track
			281 ± 6	Rb–Sr
Karkonosze Granite (51° N, 16° E)	4	25	305	K–Ar
			292	Pb–Sr
Karpacz Hornfels (51° N, 16° E)	1	7	[b]	
Kudowa Granitoid massif (51° N, 16° E)	5	37	300	K–Ar
Total	19	99		

[a] Age data after Jarmolowicz-Szulc (1986) and Puziewicz and Pin (1987)

[b] Rocks of a contact zone; the same age as for the Karkonosze Granite is assumed.

The Variscan Strzegom–Sobotka Granite (50 × 10 km) is situated close to the marginal Sudet fault zone in the northeastern part of the area (Fig. 1). The contact between the granite and country rocks indicates that the granite intruded in a magmatic state (Majerowicz, 1970b). The granites are petrographically varied, including granodiorites and alkaline granites. The rocks were emplaced at a high crustal level (Puziewicz and Pin, 1987).

The Karkonosze Granite lies to the west and consists of a number of different types of granite. The collected samples belong to the porphyritic granite, which is mainly grey in colour with a porphyritic structure and random fabric. The Karkonosze Granite is relatively rich in biotite, which is usually deformed at the edges and corroded and filled with iron ores (Szalamacha, 1970). This could imply more than one generation of ferromagnetic minerals present if these ores were formed after original emplacement and cooling of the granite.

The Karpacz Hornfels forms a thermal aureole in the micaceous schists adjacent to the Karkonosze Granite (Fig. 1). The schists were converted to an andalusite–cordierite hornfels, which was sampled near the town of Karpacz in order to compare the palaeomagnetic data from the hornfels with those from the granite.

According to Borkowska (1959) a range of petrographic varieties can be distinguished in the Kudowa Granitoid massif: monzonite granite, granodiorite and tonalite. Petrographic analysis of the contacts between the granitoid rock and the amphibole rock, together with the micaceous schists, show that they contain neither thermal nor tectonic features. On this basis it may be concluded that these rocks are not plutonic, i.e. formed by crystallization from a magma. The granitoids were formed by granitization of schists which became fused, and intruded during the final stage of their formation. In the area of the Kudowa Massif, typical vein rocks are present: aplites, porphyries and lamprophyres. The occurrence of these rocks is connected with the presence of two major dislocations striking in directions of 130° and 30° (Majerowicz, 1970a).

3. Palaeomagnetic results

Both hand samples and drilled cores were collected in the field. Sampling details and age relations are given in Table I. The natural remanent magnetization (NRM) was measured by means of a Molspin magnetometer (Sogndal) and a Jelinek JR4 (Warsaw).

The stability of NRM was determined by both thermal and alternating field (a.f.) demagnetization. Characteristic remanence components were obtained by graphical and numerical calculations using a computer line-fitting program (Kirschvink, 1980; Torsvik, 1987).

In general, the granites and granitoids have NRM intensities below 10 mA m^{-1}, and a number of specimens (~ 40%) gave intensities less than the instrumental noise level (0.2–0.5 mA m^{-1}). Thermomagnetic analyses were performed in air in a field of 600 mT. Most samples show a substantial paramagnetic contribution. The majority of samples selected for a.f. and thermal demagnetization, revealed Curie temperatures between 500 and 580 °C, indicative of nearly pure magnetite.

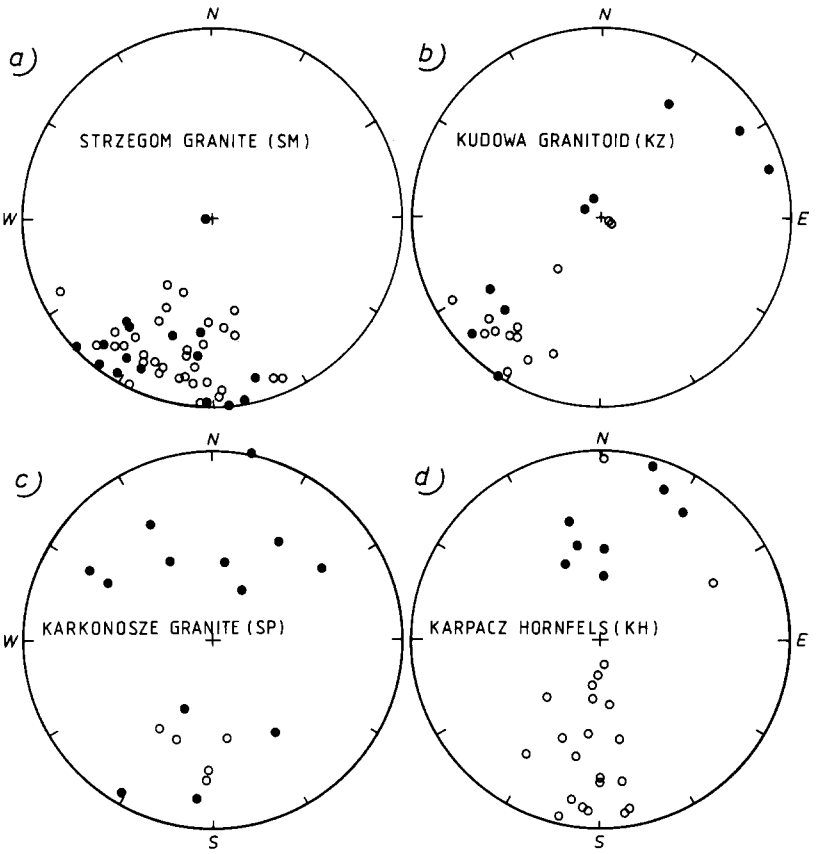


Fig. 3. Stereographic projection of stable (ChRM) NRM components obtained by thermal/a.f. demagnetization: open symbols = negative inclinations; solid symbols = positive inclinations.

Characteristic remanence components are plotted in Fig. 3. Nine sites were investigated from the Strzegom Granite complex (30 samples) revealing single-polarity magnetization with a fairly low scatter (Fig. 3a). The NRM components point south with shallow inclinations. Shallow-inclination remanence is also found in the Kudowa Granitoid (37 samples were collected from five sites, but palaeomagnetic results were only obtained from seven cores from site 1 and three cores from site 3) in addition to a few steeply dipping remanence components of both polarities (Fig. 3b).

The Karkonosze Granite was sampled in Szklarska Poreba quarry (four sites, 25 samples). Thermomagnetic analysis shows that magnetite is the dominant magnetic mineral. Thermal demagnetization reveals blocking temperatures distributed over a large range, from 225 to 560 °C. The NRM components are scattered and apparently of dual-polarity (Fig. 3c). Remanence components with shallow to intermediate negative inclinations trend south while directions with positive inclinations are mainly north trending.

Two groups of remanence components emerge from the investigation of the Karpacz Hornfels (nine hand samples collected from one exposure) (Fig. 3d); components with shallow to intermediate negative inclinations with southerly declinations, and components with northerly declinations and low to intermediate positive inclinations. The results are tentatively interpreted to represent two dual-polarity groups of remanence, one with shallow inclinations and one with intermediate inclinations.

Representative examples of thermal demagnetization are shown in Figs. 4 and 5. Specimens SM 22-A, SM 24-B (Fig. 4) and SM 21-B (Fig. 5) are

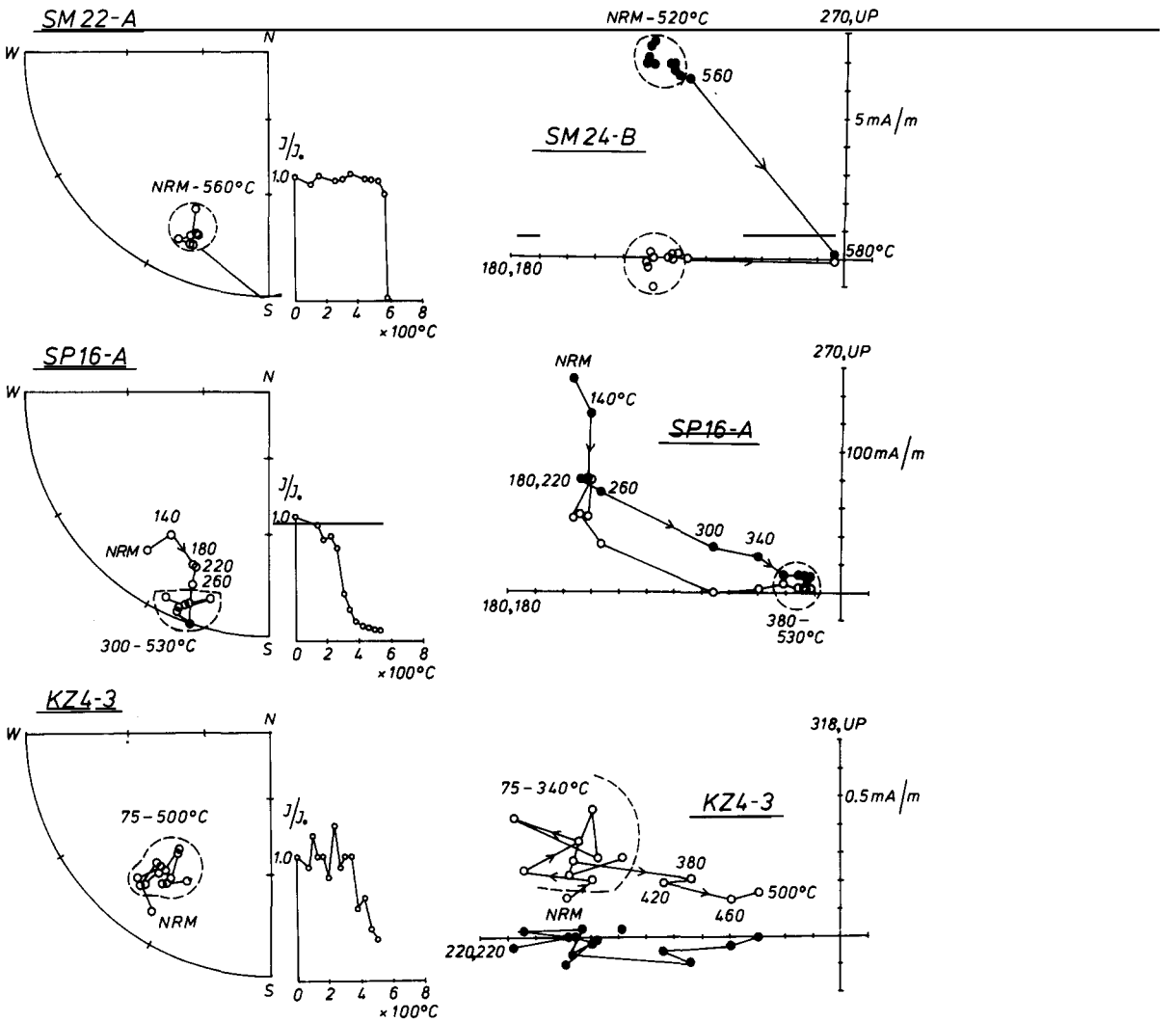


Fig. 4. Step-wise thermal demagnetization of representative specimens from the Strzegom Granite (SM 22-A and SM 24-B), the Karkonosze Granite (SP 16-A) and the Kudowa Granitoid (KZ 4-3). (left) Stereographic projections of measured directions: solid (open) symbols denote lower (upper) hemisphere; open symbols denote upper hemisphere. (middle) Intensity decay curves. (right) Orthogonal vector projections: open circles are end-point projections of demagnetization vectors in a vertical plane; solid circles in a horizontal plane.

examples of single-component directions from the Strzegom Granite, and represent the main group of NRM components found in this study. Both thermomagnetic curves and blocking temperature spectra clearly indicate that magnetite is the main carrier of the remanent magnetization. Minor changes in remanence directions and intensities occur at low to intermediate temperatures. The majority of remanence components, which have magnetization axes N–S to NE–SW and shallow inclinations, probably represent one group of characteristic directions (group A). All directions from the Strzegom Granite belong to group A, as do the majority of NRM components from the Kudowa Granitoid. The palaeomagnetic data from the Kudowa Granitoid are of lower quality than those from the Strzegom Granite, as shown by specimen KZ 4-3 (Fig. 4), and exhibit irregular

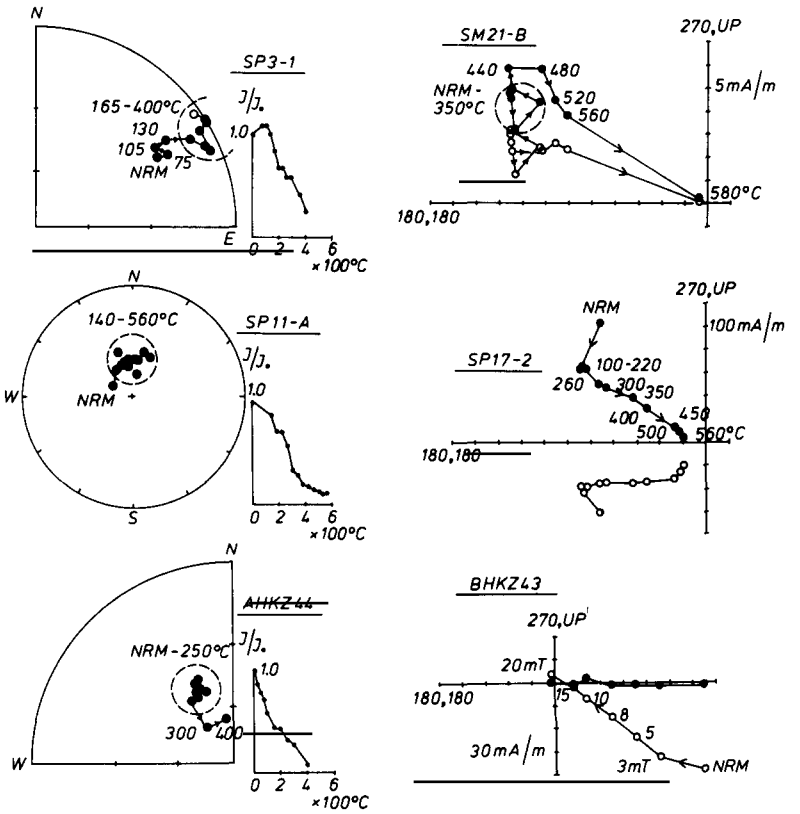


Fig. 5. Step-wise thermal/a.f. demagnetization of representative specimens from the Strzegom granite (SM 21-B), the Karkonosze granite (SP 3-1, SP 11-A and SP 17-2) and the Karpacz Hornfels (AHKZ 44 and BHKZ 43). Symbols as in Fig. 4.

intensity variations. This difference in remanence quality is attributed to the significantly lower magnetization intensity of the Kudowa samples compared with the Strzegom Granite, the remanence of the granitoid being close to the noise level of the magnetometers.

Specimens SP 16-A (Fig. 4), SP 3-1, SP 11-A and SP 17-2 (Fig. 5) typify demagnetization results from the Karkonosze Granite. The NRM components reveal dual-polarity, quite scattered magnetizations in contrast to the more tightly grouped magnetizations found in the contact aureole of the granite (specimens AHKZ 44 and BHKZ 43, Fig. 5). The remanence components from the Karkonosze Granite and the associated Karpacz Hornfels form two groups; one group of directions, which coincides with the group-A component directions, is found in the Strzegom Granite and the Kudowa granitoid (cf. specimens SP 16-A, SP 3-A and SP 17-2) and another group of directions (group B) most clearly seen in the Karpacz Hornfels (BHKZ 43 and AHKZ 44, SP 11-A).

Site-mean results are given in Table II and Fig. 6a (the Strzegom Granite, group-A component), Fig. 6b (Karpacz Hornfels, group-A component), Fig. 6c (Karkonosze Granite, group-A and group-B components) and Fig. 6d (Kappacz Hornfels, group-B component). For the Kudowa Granitoid almost all remanence directions (group A) represent a single, but laterally extensive, site. The reliability of the site-means varies considerably. High quality data are obtained from the Strzegom Granite; the results from the Karpacz Hornfels and the Kudowa Granitoid are acceptable; while the Karkonosze Granite gives quite a large scatter, as reflected by higher α_{95} values. The results from the Kudowa Granitoid define a declination $D =$ 223, while the other rocks investigated have de-

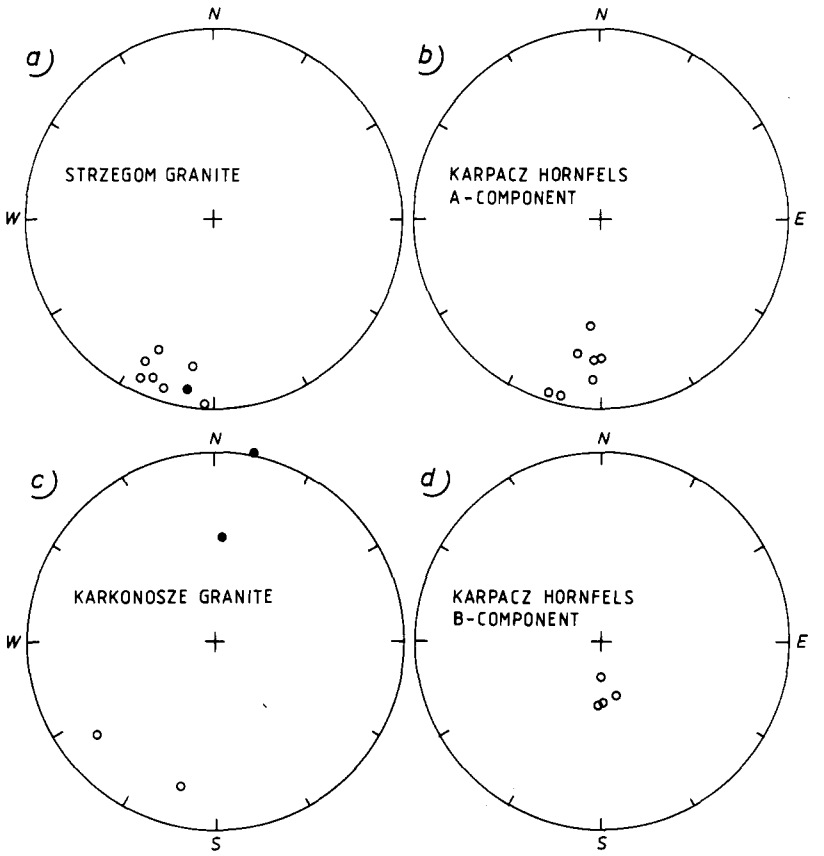


Fig. 6. Mean directions for individual sites (cf. Table II).

clinations between 177 and 201. This difference could be caused by minor rotations along the major dislocations in the Kudowa Massif (Majerowicz, 1970a). Since the NRM measurements exhibit large scatter and nearly all the results from the Kudowa Granitoid come from one site, the difference in declination is probably the result of sampling too few sites and does not have a tectonic origin.

The palaeomagnetic mean direction of the group-A component of the Sudet Granites/Granitoids remanence direction is calculated (Table III) by giving equal weight to each rock unit. This procedure seems reasonable in spite of the fact that the Strzegom Granite results are of higher reliability. The resulting overall mean is not significantly different from the results from the Strzegom Granite (Table III, $D = 198$, $I = -7$, $\alpha_{95} = 6.4$ and $N = 9$).

4. Interpretation and conclusion

The predominant magnetization in all the rock units (group-A component) yields a common mean remanence direction of $D = 202$, $I = -12$ and $\alpha_{95} = 13$ (Table III, Fig. 7).

The dual-polarity magnetic structure is suggested to represent two main groups of direction (A and B). The well-defined group-A component found in the Strzegom Granite is similar to the magnetization components in the other investigated rock units. The Strzegom Granite was intruded at a high crustal level (Puciewicz and Pin, 1987) which probably means that the intrusion

TABLE II

Palaeomagnetic site-mean results [a]

Rock formation	Site	D	I	N	Polarity [b]	α_{95}
Strzegom	1	205	−5	3		36.0
Granite	2	189	6	8		25.1
	3	201	−7	3		24.0
	4	197	−5	13		8.9
	5	206	−11	5		26.0
	6	183	−2	2		–
	7	206	−11	1		–
	8	203	−17	8		18.8
	9	188	−14	6		12.7
	Mean	198	−7	9	R	6.4
Karkonocze	1	232	−13	4		48.6
Granite	2	4	33	3		29.9
	3	194	−14	7		20.8
	4	12	0	1		–
	Mean	201	−16	4	M	21.1
Karpacz	1	193	−3	1		–
Hornfels (A)	2	183	−17	4		17.5
	3	190	−19	4		21.4
	4	186	−31	5		15.2
	5	183	−10	3		7.3
	6	197	−31	1		–
	7	180	−18	4		14.6
	Mean	188	−14	7	M	7.6
Karpacz	1	179	−54	2		
Hornfels (B)	2	183	−53	1		
	3	182	−69	2		
	4	165	−57	2		
	5	–	–	–		
	6	–	–	–		
	7	–	–	–		
	Mean	177	−58	4	M	7.4

[a] D = mean declination, I = mean inclination, N = number of specimens per site and α_{95} is calculated according to Fisher (1953).

[b] R = reversed, M = mixed.

cooled fairly quickly during a time period when the magnetic field was reversed. The predominance of reversed polarity in the group-A component is not surprising since the Kiaman superchron includes Late Carboniferous times and probably incorporates all secondary (Variscan) magnetizations. The group-B component, most clearly seen in the Karpacz Hornfels, is characterized by mixed polarity.

Low-stability components of mixed polarity and steep inclinations are present in samples from Kudowa Granite, Karkonosze Granite and Karpacz Hornfels. Regarding the Karkonosze Granite, one possible source for the low-stability components could be secondary iron ores in the altered biotite minerals. It is not possible to estimate a reliable mean direction from the demagnetization results owing to the high directional scatter. Estimated group-B directions obtained from the Karpacz Hornfels have a mean of $D = 177$, $I = -58$, $\alpha_{95} = 7.4$ and $N = 4$ (palaeomagnetic pole: 78°S, 207°E). Investigation of Upper Carboniferous volcanics from the Sudets (Westphal et al., 1987) also reveals low-stability components with steep inclinations. These low-stability components may be related to various stages of renewed activity in the Sudets during the Mesozoic phases of rifting and/or during the Alpine orogenic cycle (Malkovsky, 1987).

The investigated rock formations show radiometric ages varying from 305 to 281 Ma, i.e., Upper Carboniferous, which coincides with the probable magnetic ages. This magnetization (carried by magnetite) is probably of thermoremanent origin, related to the interval of intrusion and subsequent cooling of the granites.

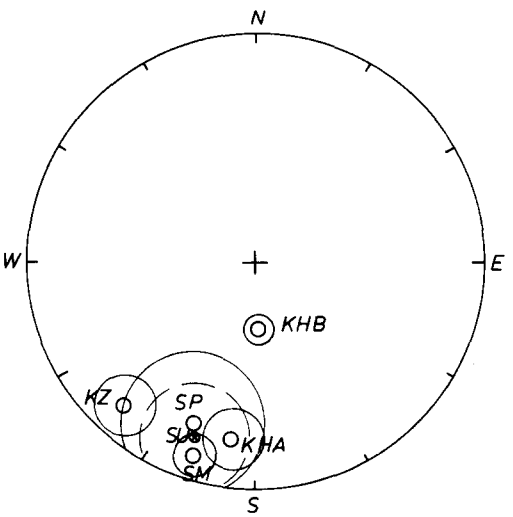


Fig. 7. Stereographic plot of mean remanence directions with statistics (α_{95}) for the Strzegom Granite (SM), Karkonosze Granite (SP), Karpacz Hornfels group-A and group-B components (KH-A and KH-B), and the Kudowa Granitoid (KZ). The overall mean remanence directions for the rocks (SU) is based on SM, SP, KH-A and KZ (cf. Table III).

TABLE III

Rock formation means and pole positions [a]

Rock formation	D	I	α_{95}	N	Polarity [b]	Calculated palaeomagnetic pole position [c]		
						S	E	dp, dm
Strzegom Granite (SM)	198	−7	6.4	9	R	40	352	3, 6
Karkonosze Granite (SP)	201	−16	21.1	4	M	44	347	11, 22
Karpacz Hornfels A (KHA)	188	−14	7.6	7	M	46	5	4, 8
Karpacz Hornfels B (KHB)	177	−58	7.4	4	M	78	207	8, 11
Kudowa Granitoid (KZ)	223	−9	8.2	18 [d]	M	31	323	4, 8
Mean Sudet granites/granitoids (SM, SP, KHA and KZ)	202	−12	13.0	4	M	42	346	7, 13

[a] D = mean declination, I = mean inclination, N = number of specimens per site, and α_{95} is calculated according to Fisher (1953).
[b] R = reversed and M = mixed.
[c] S = south, E = east, and dp, dm = semi-axis of the oval of 95% confidence about the mean pole.
[d] Mean calculated from individual specimens.

A comparison of the calculated palaeomagnetic pole (VGP) position of 42°S, 346°E with the reference polar wander path for northwest Europe, reveals that the Sudet pole of Upper Carboniferous age fits the polar wander path well (Fig. 8). Thus, it is concluded that no major tectonic movement has taken place in the Sudets relative to stable Europe since the Upper Carboniferous. The

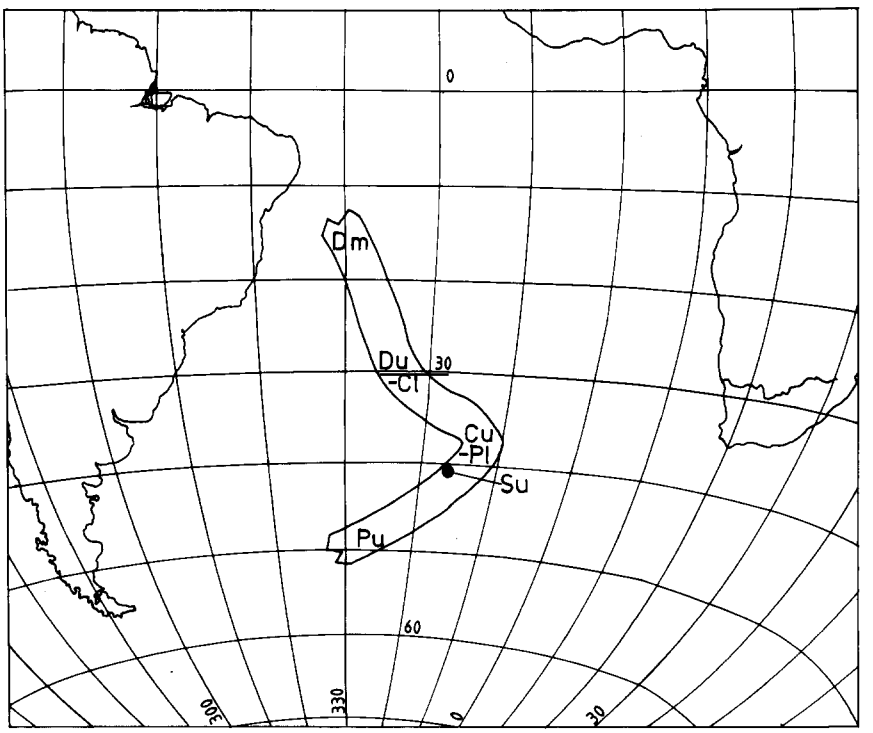


Fig. 8. The palaeopoles for the Strzegom and Karkonosze Granites, the Karpacz Hornfels and the Kudowa Granitoid (SU), are plotted on the APWP for the Eurasian plate from Middle Devonian to Upper Permian as proposed by Perroud (1986) and Tarling (1983).

results obtained in the present study are also in good agreement with data from a study of Upper Carboniferous volcanics from the Sudets (Westphal et al., 1987) which yield a primary magnetization component ($D = 192$, $I = -2$, $\alpha_{95} = 11$, $N = 8$) with a pole at 39°S, 1°E.

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